The Sharma Absolute Boundary Summation (ABS) Model: A Bounded, Sign-Preserving Multiplicative Noise Factor for Embedded Simulation

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Abstract

We study the Sharma Absolute Boundary Summation (ABS) model, a multiplicative noise factor $\eta = N/(N+S)$, where S is a sum of n independent uniform integers on $[a, b]$ and $N = n(|a| + |b|)$. The factor is applied to a deterministic signal as $v = v_{\text{ideal}}\,[1 + \alpha(\eta - 1)]$. We prove that the denominator can never vanish when $b \neq 0$, that η is strictly positive and bounded (in $[2/3, 2]$ for symmetric bounds), and that the multiplier preserves the sign of the signal for $0 \leq \alpha \leq 1$. We give the exact distribution of η as a finite sum, closed-form approximations of its mean and variance, and show that η is the reciprocal transform of a scaled Irwin–Hall variate. A practical consequence is that the noise amplitude is set by n and α, and is almost independent of the bounds $[a, b]$, which only set the quantization. η has a positive bias, $\mathbb{E}[\eta] > 1$, which we quantify and remove by an exact normalization. We provide floating-point and integer-only (Q16.16) implementations and reproducible experiments, and state the model's limitations: it produces white, thin-tailed noise, and it does not replace coherent noise or heavy-tailed financial models.

1 Introduction

Testing embedded firmware and simulations often requires perturbing a deterministic signal (a sensor reading, a velocity, a timing value) with noise whose size scales with the signal. Multiplicative noise of the form $v = v_{\text{ideal}}(1 + \epsilon)$ is the standard choice for this [3, 7]. When ϵ is Gaussian, generating it needs either transcendental functions (Box–Muller [1]) or a sum of uniforms (the Irwin–Hall approximation [5, 4]), and the factor $1 + \epsilon$ can become zero or negative, which flips the sign of the signal.

The ABS model builds the factor from a bounded integer sum and a single division. This paper makes the following contributions:

1. proofs that the factor is always defined, strictly positive and bounded, and that the perturbed signal keeps its sign (Section 3);
2. the exact distribution of η, closed-form moments, and its relation to the Irwin–Hall distribution (Sections 4–5);
3. design rules for n and α, and an exact bias correction (Section 6);
4. float and integer-only implementations with an unbiased integer sampler, and reproducible experiments (Sections 7–8).

We aim for a small, well-characterized engineering tool, and we state where it should not be used (Section 9).

2 Definition

Definition 1 (Sharma ABS factor)**.** Let $a < b$ be integers, $n \geq 1$, and $r_1, \ldots, r_n$ independent and uniform on $\{a, a+1, \ldots, b\}$. Define

$$W = |a| + |b|, \qquad N = nW, \qquad S = \sum_{i=1}^{n} r_i, \qquad \eta = \frac{N}{N+S}. \tag{1}$$

*This version supersedes the draft of February 2026. The quantization result reported there (the "Magic 11" effect) was an artifact of histogram binning; it is corrected in Section 8.2. The statistical table of the earlier draft is replaced by exact values with stated parameters.

For an influence coefficient $\alpha \geq 0$, the perturbed signal is

$$v_{\text{real}} = v_{\text{ideal}} \left[1 + \alpha(\eta - 1)\right] = v_{\text{ideal}} \left[(1 - \alpha) + \alpha\eta\right]. \tag{2}$$

We call n the *complexity* and $[a, b]$ the *bounds*. Throughout, $m = b - a + 1$ is the number of values each r_i can take.

3 Basic properties

Proposition 1 (No division by zero)**.** *For every outcome, $N + S \geq n(W + a)$. Moreover $W + a > 0$ if and only if $b \neq 0$. Hence the denominator of* (1) *is strictly positive for every outcome if and only if $b \neq 0$.*

Proof. Each $r_i \geq a$, so $S \geq na$ and $N + S \geq n(W + a)$, with equality when all $r_i = a$, an outcome of positive probability. If $a \geq 0$ then $b > a \geq 0$ and $W + a = 2a + b > 0$. If $a < 0$ then $W + a = |b|$, which is positive iff $b \neq 0$. □

So the "stability guard" of the earlier draft (returning 1 when the denominator is zero) is only reachable when $b = 0$. Implementations should reject $b = 0$ instead.

Corollary 1 (Bounds and positivity)**.** *If $b \neq 0$, η is strictly positive and*

$$\frac{W}{W + b} \;\leq\; \eta \;\leq\; \frac{W}{W + a}. \tag{3}$$

For symmetric bounds $[-c, c]$, $c \geq 1$, this is $2/3 \leq \eta \leq 2$ for every c and n.

Proof. η is decreasing in S, and $na \leq S \leq nb$; substitute both ends. Both ends are positive by Proposition 1. □

Corollary 2 (Sign preservation)**.** *If $b \neq 0$ and $0 \leq \alpha \leq 1$, then $(1 - \alpha) + \alpha\eta > 0$, so v_{real} has the same sign as v_{ideal}, and $v_{real} = 0$ only when $v_{ideal} = 0$.*

This is the precise version of the claim that ABS "respects physical limits": a non-negative quantity (a voltage on a 0–5 V rail, a speed, a price) stays non-negative. It does *not* respect an upper limit: with $\alpha = 1$ and symmetric bounds, the signal can be multiplied by up to 2 (Section 9).

4 Exact distribution

Proposition 2 (States)**.** *η takes exactly $n(b - a) + 1$ distinct values, one for each integer $s \in [na, nb]$, with*

$$P\Big(\eta = \tfrac{N}{N+s}\Big) = P(S = s) = m^{-n} \sum_{j \geq 0} (-1)^j \binom{n}{j} \binom{k - jm + n - 1}{n - 1}, \qquad k = s - na, \tag{4}$$

where binomials with negative upper index are zero. In T independent draws, the expected number of distinct values observed is

$$D(T) = \sum_{s=na}^{nb} \Big[1 - \big(1 - P(S = s)\big)^T\Big]. \tag{5}$$

Proof. $S - na$ is a sum of n independent uniforms on $\{0, \ldots, m - 1\}$; (4) is the standard inclusion–exclusion count of such compositions. The map $s \mapsto N/(N + s)$ is strictly decreasing, so distinct sums give distinct η. Equation (5) follows from linearity of expectation over the indicator that state s is seen. □

Because the support is finite, every moment of η is a finite sum, $\mathbb{E}[\eta^k] = \sum_s P(S = s)\,(N/(N + s))^k$, which can be evaluated exactly on a host computer (in practice by convolving n uniform PMFs). We use these exact values as ground truth.

5 Moments, and the role of the bounds

Write $\eta = 1/(1 + X)$ with $X = S/N$. Then

$$\mu_X = \mathbb{E}[X] = \frac{a + b}{2W}, \qquad \sigma_X^2 = \operatorname{Var}[X] = \frac{m^2 - 1}{12\,nW^2}. \tag{6}$$

A second-order expansion of $1/(1 + X)$ about μ_X (the delta method [2]) gives

$$\mathbb{E}[\eta] \approx 1 - \mu_X + \mu_X^2 + \sigma_X^2, \qquad \operatorname{sd}(\eta) \approx \frac{\sigma_X}{(1 + \mu_X)^2}. \tag{7}$$

Symmetric bounds. For $[a, b] = [-c, c]$, $\mu_X = 0$ and $\sigma_X^2 = (c+1)/(12nc)$, so

$$\mathrm{sd}(\eta) \approx \sqrt{\frac{1+1/c}{12\,n}} \xrightarrow{c\to\infty} \frac{1}{\sqrt{12n}}, \qquad \mathbb{E}[\eta] - 1 \approx \frac{1+1/c}{12\,n} > 0. \tag{8}$$

Two consequences follow. First, the noise amplitude is controlled by n, while c only matters through the factor $1 + 1/c$, which is between 1 and 2. Bounds of $[-10, 10]$ and $[-1000, 1000]$ produce almost the same noise (Table 1, Figure 2); larger bounds only make the quantization finer (Proposition 2). Second, $\mathbb{E}[\eta] > 1$. This is Jensen's inequality [6]: $x \mapsto 1/(1+x)$ is strictly convex, so $\mathbb{E}[\eta] > 1/(1 + \mathbb{E}[X]) = 1$ whenever X is non-degenerate and $\mathbb{E}[X] = 0$. With asymmetric bounds the bias is larger because $\mu_X \neq 0$; for $[-12, 11]$, $\mu_X = -1/46$ and $\mathbb{E}[\eta] = 1.0305$.

Relation to Irwin–Hall. As $c \to \infty$, r_i/c tends to a continuous uniform variable on $[-1, 1]$, so $X \to (\sum_{i=1}^{n} U_i - n/2)/n$ with U_i uniform on $[0, 1]$. Writing Z_n for the standardized Irwin–Hall sum [5, 4],

$$\eta \;\approx\; \frac{1}{1 + Z_n/\sqrt{12n}}. \tag{9}$$

For $n = 12$ this is $1/(1+Z_{12}/12)$, where $Z_{12} = \sum U_i - 6$ is the classical "sum of twelve uniforms" approximation to a standard normal variable [8]. ABS is therefore close to standard multiplicative near-Gaussian noise, $1 - Z/\sqrt{12n}$, to first order. The reciprocal adds a positive skew and keeps the factor positive and bounded, which the plain Gaussian form does not.

6 Design rules

Amplitude. By (2), the multiplier $M = 1 + \alpha(\eta - 1)$ has $\mathrm{sd}(M) = \alpha\,\mathrm{sd}(\eta)$ and $\mathbb{E}|M - 1| = \alpha\,\mathbb{E}|\eta - 1|$: every deviation statistic is *exactly* linear in α (Table 3). Since n costs n random draws per sample and α costs nothing, we recommend choosing n for the shape (larger n means less skew and less bias) and then setting

$$\alpha = \frac{\sigma_{\mathrm{target}}}{\mathrm{sd}(\eta)} \approx \sigma_{\mathrm{target}} \sqrt{\frac{12n}{1+1/c}}. \tag{10}$$

For example, $\sigma_{\mathrm{target}} = 2\%$ with $n = 12$, $c = 10$ gives $\alpha \approx 0.22$. Keeping $\alpha \leq 1$ preserves the sign guarantee; this limits the amplitude to $\mathrm{sd}(\eta)$.

Bias correction. If a mean multiplier of exactly one matters, use $\tilde{\eta} = \eta/\mathbb{E}[\eta]$ with $\mathbb{E}[\eta]$ computed exactly once on the host (Section 4). Then $\mathbb{E}[\tilde{\eta}] = 1$, $\mathbb{E}[M] = 1$, and positivity is unchanged. For $[-10, 10]$, $n = 12$, the constant is 1.0078.

Bounds. Use symmetric bounds unless a bias is wanted. Choose c only for resolution: the factor has $2nc + 1$ levels, which matters only for very small c (e.g. $c = 1$, $n = 12$ gives 25 levels).

7 Implementation

The reference C implementation is a single header with no dependencies beyond `stdint.h`. It uses:

- a seeded xorshift32 generator [11], with a re-entrant variant that takes the state as an argument (for ISRs and threads);
- Lemire's multiply-shift method with rejection [9] to draw unbiased integers on $[a, b]$. The earlier `rand() % m` has modulo bias and was never seeded;
- an integer-only path that returns η in Q16.16 format. When $N < 2^{16}$ it needs one 32-bit unsigned division, which is a single instruction on Cortex-M3/M4/M7. Otherwise it falls back to a 64-bit division;
- input validation that rejects $n \leq 0$ and $b = 0$ (Proposition 1).

With $|a|, |b|, n \leq 2^{15} - 1$ every intermediate fits in a signed 32-bit integer. On our test host the Q16 result differs from the float result by at most one Q16 step (1.5×10^{-5}) over 2×10^6 draws.

8 Experiments

All numbers below are produced by `analysis/make_paper_results.py` (seed 2026) and `analysis/bench.c` in the repository.

8.1 Moments: exact, approximate and simulated

Table 1 compares exact moments from (4), the approximations (7), and 10^6 Monte Carlo draws. The exact and simulated values agree to within sampling error. The approximation of the mean is within 0.002, and the approximation of the standard deviation is within 9% for small n and within 1% for $n \geq 48$.

Table 1: Moments of η. "Ex." exact, "Appr." Eq. (7), "MC" 10^6 draws.

			$\mathbb{E}[\eta]$			$\mathrm{sd}(\eta)$			skew
Bounds	n	states	Ex.	Appr.	MC	Ex.	Appr.	MC	Ex.
$[-1, 1]$	12	25	1.0145	1.0139	1.0146	0.1247	0.1179	0.1246	0.73
$[-10, 10]$	4	81	1.0245	1.0229	1.0247	0.1654	0.1514	0.1654	0.88
$[-10, 10]$	12	241	1.0078	1.0076	1.0079	0.0901	0.0874	0.0903	0.52
$[-10, 10]$	48	961	1.0019	1.0019	1.0019	0.0440	0.0437	0.0441	0.26
$[-1000, 1000]$	12	24001	1.0071	1.0070	1.0070	0.0857	0.0834	0.0857	0.50
$[-12, 11]$	12	277	1.0305	1.0298	1.0305	0.0937	0.0908	0.0936	0.53
$[-50, 50]$	50	5001	1.0017	1.0017	1.0017	0.0415	0.0412	0.0415	0.25

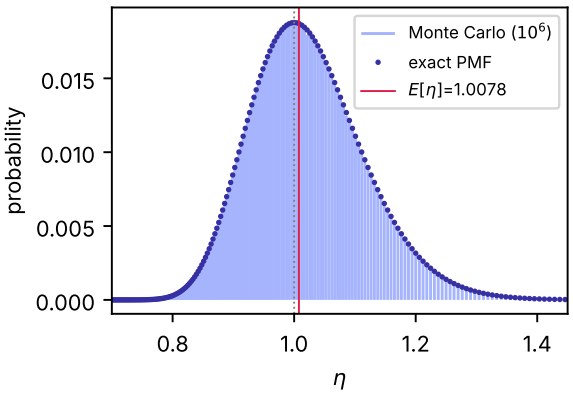


Figure 1: Exact distribution of η for $[-10, 10]$, $n = 12$, against 10^6 draws. The distribution is right-skewed and its mean exceeds 1.

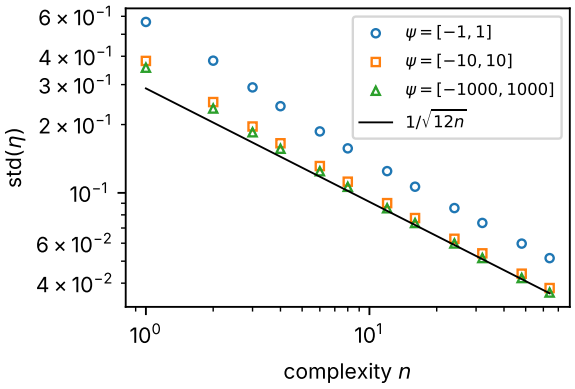


Figure 2: $\mathrm{sd}(\eta)$ against n for three bounds. The spread follows $1/\sqrt{12n}$ and barely depends on the bounds.

8.2 Number of distinct values

The earlier draft reported only 11 distinct values of η in 100 trials for $n = 12$, $[-12, 11]$, and 88 in 1000 trials for $n = 50$, $[-50, 50]$. Proposition 2 shows that this is not possible: these settings have 277 and 5001 states, and the expected counts are about 61 and 566. Table 2 confirms this over 20 independent runs. The earlier figure of 11 most likely counted the occupied bins of a histogram of width 0.02 over $[0.9, 1.1]$. There is no special bottleneck at $n = 12$; the number of states is always $n(b - a) + 1$.

Table 2: Distinct values of η: theoretical states, expected count $D(T)$ from (5), and observed mean (range) over 20 runs.

Bounds	n	T	states	$D(T)$	observed
$[-12, 11]$	12	100	277	60.8	59.5 (51–65)
$[-12, 11]$	12	10000	277	156.6	155.4 (151–160)
$[-50, 50]$	50	1000	5001	565.7	564.6 (550–581)

8.3 Effect of α

Table 3 gives exact statistics of the multiplier M for $[-10, 10]$, $n = 12$. The mean absolute deviation and the 99.9% two-sided deviation bound are exactly proportional to α. The mean of M is $1 + \alpha(\mathbb{E}[\eta] - 1)$, which shows the bias when no correction is applied.

Table 3: Exact statistics of $M = 1 + \alpha(\eta - 1)$, bounds $[-10, 10]$, $n = 12$, no bias correction.

α	$\mathbb{E}[M]$	$\mathbb{E}\lvert M - 1\rvert$ (%)	99.9% bound on $\lvert M - 1\rvert$ (%)
0.00	1.0000	0.00	0.00
0.25	1.0020	1.78	9.68
0.50	1.0039	3.56	19.36
1.00	1.0078	7.11	38.73

8.4 Cost

Table 4 reports throughput on a desktop-class host (Intel Xeon at 2.1 GHz, GCC 13.3, `-O2`, 2×10^7 samples, median of 3 runs, the same xorshift32 generator for all methods). The cost of ABS is dominated by its n random draws: it matches Irwin–Hall with the same n, and it is slower than Box–Muller on a machine with a hardware FPU and fast `libm`. The case for ABS on microcontrollers rests on its operation count: integer additions and a single integer division, with no transcendental functions, which are expensive on cores without an FPU. We have not yet measured cycle counts on microcontroller targets; this is left for future work.

Table 4: Host throughput (lower is faster). Not representative of microcontrollers.

Method	Operations per sample	ns/sample
ABS, float ($n = 12$)	12 RNG + 12 add + 1 float div	31.4
ABS, Q16.16 ($n = 12$)	12 RNG + 12 add + 1 int div	31.0
Irwin–Hall ($n = 12$)	12 RNG + 12 add	30.1
Box–Muller (per sample)	1 RNG + $\frac{1}{2}$(log, $\sqrt{}$, sin, cos)	12.0

8.5 Illustrations

Figure 3 applies the bias-corrected factor to two signals. On the left, a sensor voltage between 0.1 and 4.9 V is perturbed with $n = 48$, $\alpha = 0.5$. Near 0 V the noise shrinks with the signal and the reading never goes negative. Near the top of the range, however, the noisy signal exceeds the 5 V rail in 199 of 200 independent runs, and goes below 0 V in none. Users who need an upper limit must clip it explicitly. On the right, a damped oscillation is perturbed with $n = 12$; the noise decays with the envelope, as expected from multiplicative noise.

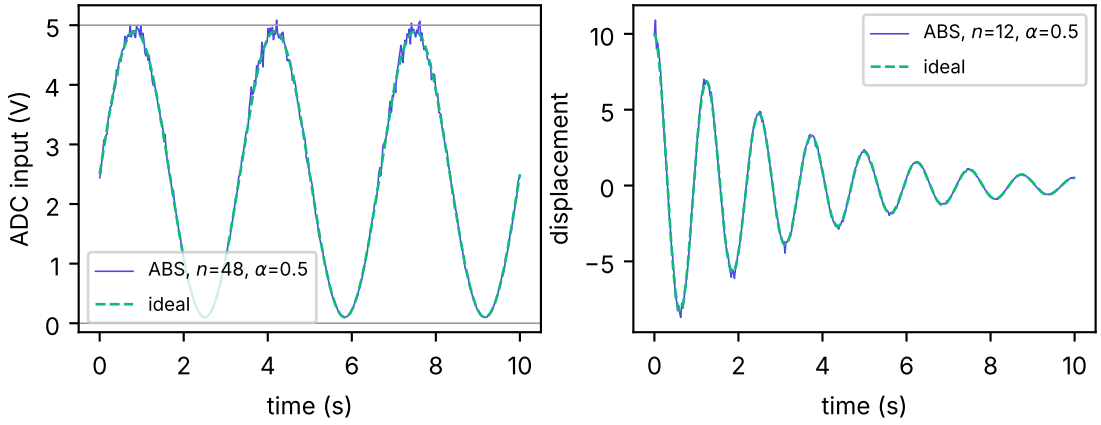


Figure 3: Left: ADC input on a 0–5 V rail; the lower rail is respected, the upper one is not. Right: damped oscillation. Bias-corrected η in both.

9 Limitations and scope

- **White noise only.** Successive values of η are independent. ABS is not a substitute for coherent noise such as Perlin noise [12], which is correlated across space or time; correlated jitter needs filtering or a different generator.
- **Thin, bounded tails.** $\eta \in [2/3, 2]$ for symmetric bounds and its tails are lighter than Gaussian. It is not suitable for heavy-tailed phenomena such as large financial market moves [10].
- **Upper limits are not respected.** Positivity holds, but the multiplier can reach $1 + \alpha$.
- **Bias and skew.** $\mathbb{E}[\eta] > 1$ and the distribution is right-skewed, especially for small n. The bias is removable (Section 6); the skew is not.
- **Randomness quality.** The statistical guarantees assume independent uniform draws; xorshift32 is adequate for simulation but is not cryptographic.

10 Conclusion

The Sharma ABS factor is a simple multiplicative noise generator with provable guarantees: it is always defined when $b \neq 0$, strictly positive, bounded, and preserves the sign of the signal for $\alpha \leq 1$. Its exact distribution is a finite sum, its amplitude is governed by n and α rather than by the bounds, and its positive bias can be removed exactly. To first order it behaves like Irwin–Hall multiplicative noise. Its value lies in combining these guarantees with integer-only arithmetic, rather than in raw speed on hosts with an FPU. Future work is to measure cycle counts on Cortex-M targets and to add a correlated (filtered) variant.

Code and data availability

Source code (C header, Python, CLI) and all scripts that generate the tables and figures are available under the MIT License at `https://github.com/kumarnitish378/sharma-abs-model`.

How to cite

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References

[1] G. E. P. Box and M. E. Muller, "A note on the generation of random normal deviates," *Ann. Math. Statist.*, 29(2):610–611, 1958.

[2] G. Casella and R. L. Berger, *Statistical Inference*, 2nd ed., Duxbury, 2002.

[3] C. W. Gardiner, *Handbook of Stochastic Methods*, Springer, 1985.

[4] P. Hall, "The distribution of means for samples of size N drawn from a population in which the variate takes values between 0 and 1, all such values being equally probable," *Biometrika*, 19(3/4):240–245, 1927.

[5] J. O. Irwin, "On the frequency distribution of the means of samples from a population having any law of frequency with finite moments, with special reference to Pearson's Type II," *Biometrika*, 19(3/4):225–239, 1927.

[6] J. L. W. V. Jensen, "Sur les fonctions convexes et les inégalités entre les valeurs moyennes," *Acta Math.*, 30:175–193, 1906.

[7] P. E. Kloeden and E. Platen, *Numerical Solution of Stochastic Differential Equations*, Springer, 1992.

[8] D. E. Knuth, *The Art of Computer Programming, Vol. 2: Seminumerical Algorithms*, 3rd ed., Addison-Wesley, 1997.

[9] D. Lemire, "Fast random integer generation in an interval," *ACM Trans. Model. Comput. Simul.*, 29(1):3, 2019.

[10] B. Mandelbrot, "The variation of certain speculative prices," *J. Business*, 36(4):394–419, 1963.

[11] G. Marsaglia, "Xorshift RNGs," *J. Stat. Softw.*, 8(14), 2003.

[12] K. Perlin, "An image synthesizer," *SIGGRAPH Comput. Graph.*, 19(3):287–296, 1985.